00.Parameter Expansion

Sunday, March 15, 2026 3:38 PM

"${var^}" capitalize the first letter

"${var^^}" capitalize the entire string

"${var^^d}" capitalize all the d's

"${var^d}" capitalize the first d

"${var^[da]}" pattern matching capitalization

"${var,}" same as ^ but for lower casing

cd go to home directory (no args)

var = ${1:-value} if not set, set it to value

var = ${1?} parameter required - accepts an empty str

var = ${1?'message'} provides an error message

var = ${1:?} does not accept an empty string

${var /char /char} char that replaces it / once
↳ char to look for
${var // char / char} replaces all

dirname $var - gives the directory name
basename $var - gives the file name

${var # */} from the beginning of the line delete anything up to the first '/'
${var ## */} from the beginning of the line delete anything up to the last '/'

% opposite of # (start at the end of the line)

& back reference the matched character

${var :n} make a substring from n^{th} char
${var :n:m} m → how many chars
${var: -n} start from the end

${var @ A} shell safe way - can be parsed

${var@a} attribute
${var@Q} quote it
${var@U} uppercase

} parameter transformation